

ML Experiment no 6 - KNN Classification

Import Libraries

```
In [1]: import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
In [2]: url = "https://raw.githubusercontent.com/uiuc-cse/data-fa14/gh-pages/data/iris.csv"
data = pd.read_csv(url)
```

```
In [3]: data.head()
```

```
Out[3]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
In [5]: data.size
```

```
Out[5]: 750
```

```
In [6]: data.columns
```

```
Out[6]: Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
              'species'],
              dtype='object')
```

```
In [7]: data.describe
```

```
Out[7]: <bound method NDFrame.describe of
0      5.1      3.5      1.4      0.2      setosa
1      4.9      3.0      1.4      0.2      setosa
2      4.7      3.2      1.3      0.2      setosa
3      4.6      3.1      1.5      0.2      setosa
4      5.0      3.6      1.4      0.2      setosa
..      ...      ...      ...      ...      ...
145     6.7      3.0      5.2      2.3     virginica
146     6.3      2.5      5.0      1.9     virginica
147     6.5      3.0      5.2      2.0     virginica
148     6.2      3.4      5.4      2.3     virginica
149     5.9      3.0      5.1      1.8     virginica
```

```
[150 rows x 5 columns]>
```

```
In [9]: data.tail()
```

```
Out[9]:
```

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

```
In [10]: X = data.drop('species', axis=1)
y = data['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
```

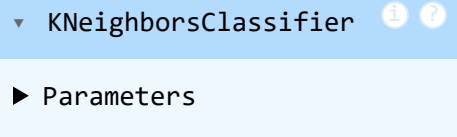
```
In [13]: scaler = StandardScaler()
```

```
In [15]: X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.fit_transform(X_test)
```

```
In [16]: knn_model = KNeighborsClassifier(n_neighbors = 5)
```

```
In [18]: knn_model.fit(X_train_scaled, y_train)
```

```
Out[18]:
```



```
In [21]: y_predict_knn = knn_model.predict(X_test_scaled)
```

```
In [29]: print("\nClassification Result:")
print("Accuracy:", round(accuracy_score(y_test, y_predict_knn)*100, 2), "%")
print(confusion_matrix(y_test, y_predict_knn))
print(classification_report(y_test, y_predict_knn))
```

Classification Result:

Accuracy: 96.67 %

```
[[10  0  0]
 [ 0  9  0]
 [ 0  1 10]]
```

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	0.90	1.00	0.95	9
virginica	1.00	0.91	0.95	11
accuracy			0.97	30
macro avg	0.97	0.97	0.97	30
weighted avg	0.97	0.97	0.97	30

```
In [ ]:
```